Exercise 1

We would like to print the following group of geometries combining different pattern types. The patterns should be combined in single G-code file.

The description of the geometries is the following:

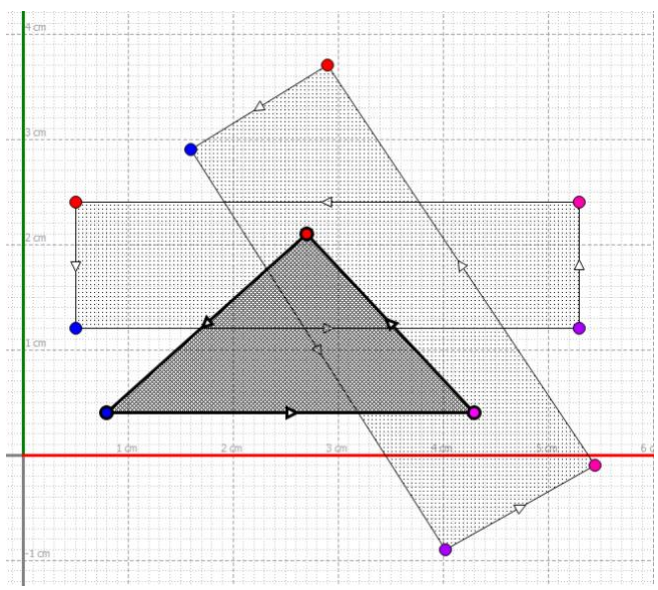
- "Rectangle 1":[[5, 12], [53, 12],[53, 24],[5, 24]]
- "Rectangle 2":[[16, 29],[40.213, -9],[54.483, -1],[29, 37]]
- "Triangle":[[8, 4],[43, 4],[27, 21]

The pattern assigned to each part should be:

- Rectilinear pattern for rectangle 1 (0 -degree rotation)
- Honeycomb pattern for honeycomb for rectangle 2
- Spiral for triangle
- Rectilineal pattern with 90 -degree rotation for the intersection between rectangle 1 and rectangle 2
- Rectilinear pattern with 120 -degree rotation for the intersection between rectangle 1 and the triangle.
- No pattern should be assigned to the intersection between rectangle 2 and the triangle.
- Another pattern should be assigned to the intersection between the 3 shapes.

In order to do this, the ClipperLib library should be used either with Python or C++ (https://www.angusj.com/clipper2/Docs/Overview.htm)

Please send back your code as your response to this test.



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Exercise 2

A rectangular template with dimensions W x H is defined in a 2D cartesian plane where $0 < x < W$ and $0 < y < H$. You are required to perform t**wo distinct spatial mappings** to transform a rectilinear pattern etched on this template:

- Part A (2D Circular Mapping): Derive the coordinate transformation equations to map the rectangle onto a 2D ring with an inner radius. In this mapping, the x-axis of the pattern must correspond to the angular displacement $\theta \in [0,2\pi]$, and the y-axis must correspond to the radial distance. Calculate the resulting Cartesian coordinates (x',y') for the four corners of the rectangle.
- Part B (3D Cylindrical Mapping): Apply a transformation to "wrap" the same rectangular template around a vertical cylinder of radius. The x-axis of the rectangle must represent the arc length along the cylinder's circumference, while the y-axis represents the vertical height z. Provide the general transformation function and determine the 3D coordinates of the corner points if $W=2\pi R$.

Calculate the pattern for the 2 cases and represent them in a plot.

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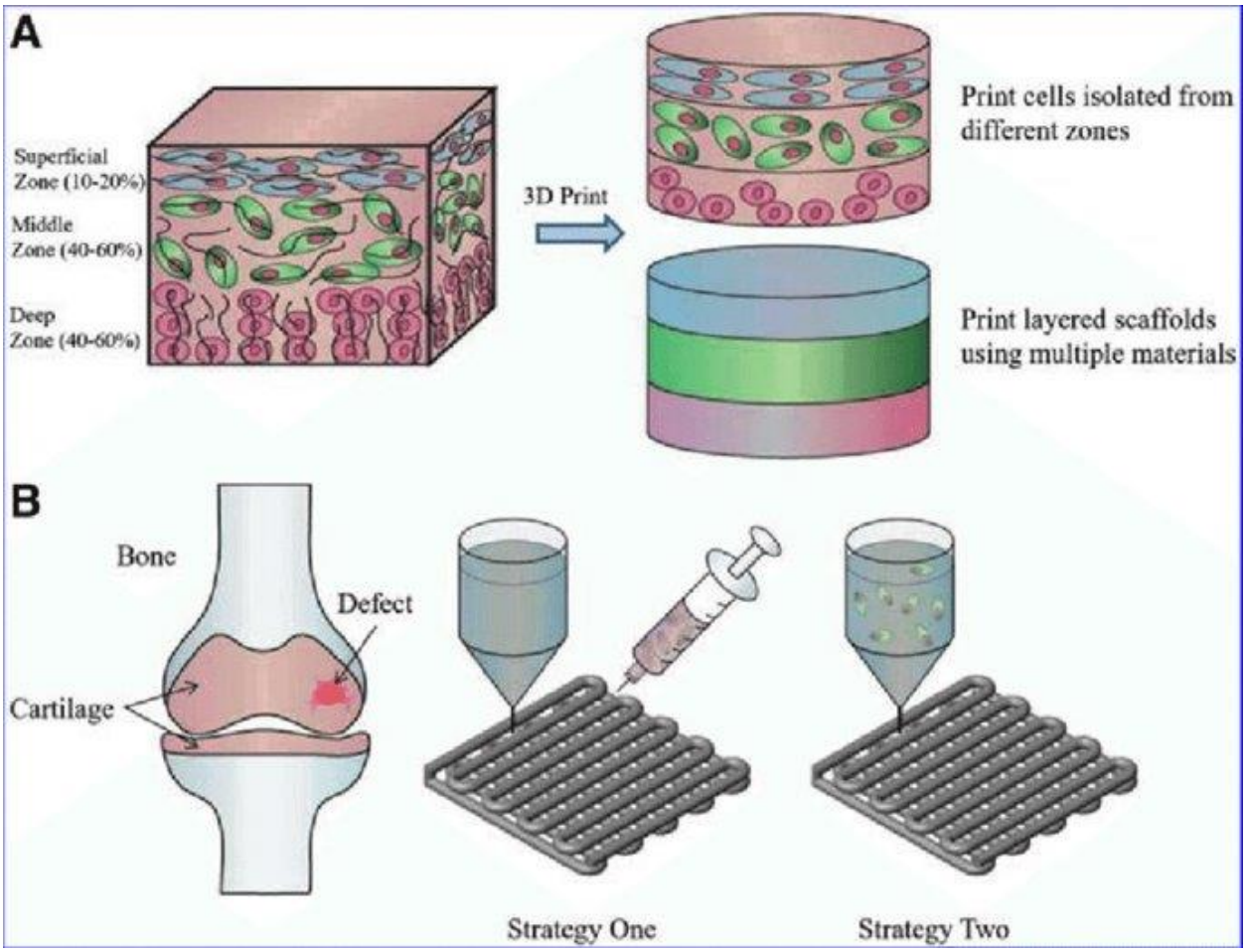
Exercise 3

Articular cartilage is a non-homogeneous tissue composed of three distinct zones: superficial, middle, and deep, which vary significantly in their cellular organization, collagen fiber orientation, and mechanical stiffness. Your task is to propose a design for a 3D biomodel that replicates these three layers. You must take into consideration the specific biomechanical properties (such as Young's Modulus and permeability) for each zone to ensure the model functions replicates real tissue. In your response, explain the manufacturing techniques you would use to create these distinct layers, how you would ensure a seamless interface between them to prevent delamination, and which professional profiles you would collaborate with to validate the model's performance against human physiological data.